

# Particle Physics 1 Basic Concepts: Lecture 9 : New Revolutions in Particle Physics: Basic Concepts

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## Chapter 1

# Lagrangians, Fields, and Local Particle Processes

Boson fields, fermion fields, and the electromagnetic field all obey wave-like equations. Those equations involve the fields themselves together with their space and time derivatives. One could list a separate equation of motion for every field, but a Lagrangian collects the entire system into one compact expression. Applying the Euler–Lagrange rule to that expression produces the equations for all the fields at once.

This construction is closely related to stationary action. Here the immediate task is more concrete: identify what a field Lagrangian depends on, use it to obtain a few classical equations, and then reinterpret the same field expressions quantum mechanically. In the quantum theory the Lagrangian governs particle motion and interaction, and ultimately the amplitudes and cross sections for collisions.

### 1.1 A Compact Code for Field Equations

For one scalar field, the Lagrangian density depends on the field and on its first derivatives,

$$\mathcal{L} = \mathcal{L}(\phi, \partial_t \phi, \partial_x \phi, \dots) = \mathcal{L}(\phi, \partial_\mu \phi). \quad (1.1)$$

The explicit list of time and spatial derivatives is often more useful while calculating; the notation  $\partial_\mu \phi$  packages the same list in relativistic form.

The field Euler–Lagrange equation is

$$\frac{\partial}{\partial t} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} + \frac{\partial}{\partial x} \frac{\partial \mathcal{L}}{\partial \phi_x} + \frac{\partial}{\partial y} \frac{\partial \mathcal{L}}{\partial \phi_y} + \frac{\partial}{\partial z} \frac{\partial \mathcal{L}}{\partial \phi_z} = \frac{\partial \mathcal{L}}{\partial \phi}, \quad (1.2)$$

where

$$\dot{\phi} = \frac{\partial \phi}{\partial t}, \quad \phi_x = \frac{\partial \phi}{\partial x}, \quad \phi_y = \frac{\partial \phi}{\partial y}, \quad \phi_z = \frac{\partial \phi}{\partial z}. \quad (1.3)$$

If  $\mathcal{L}$  contains several fields, this rule is applied once for each field. The result is the system of equations of motion.

The simplest scalar example is

$$\mathcal{L} = \frac{1}{2} \dot{\phi}^2 - \frac{1}{2} \phi_x^2 - \frac{1}{2} \phi_y^2 - \frac{1}{2} \phi_z^2 - V(\phi) = \frac{1}{2} \dot{\phi}^2 - \frac{1}{2} (\nabla \phi)^2 - V(\phi). \quad (1.4)$$

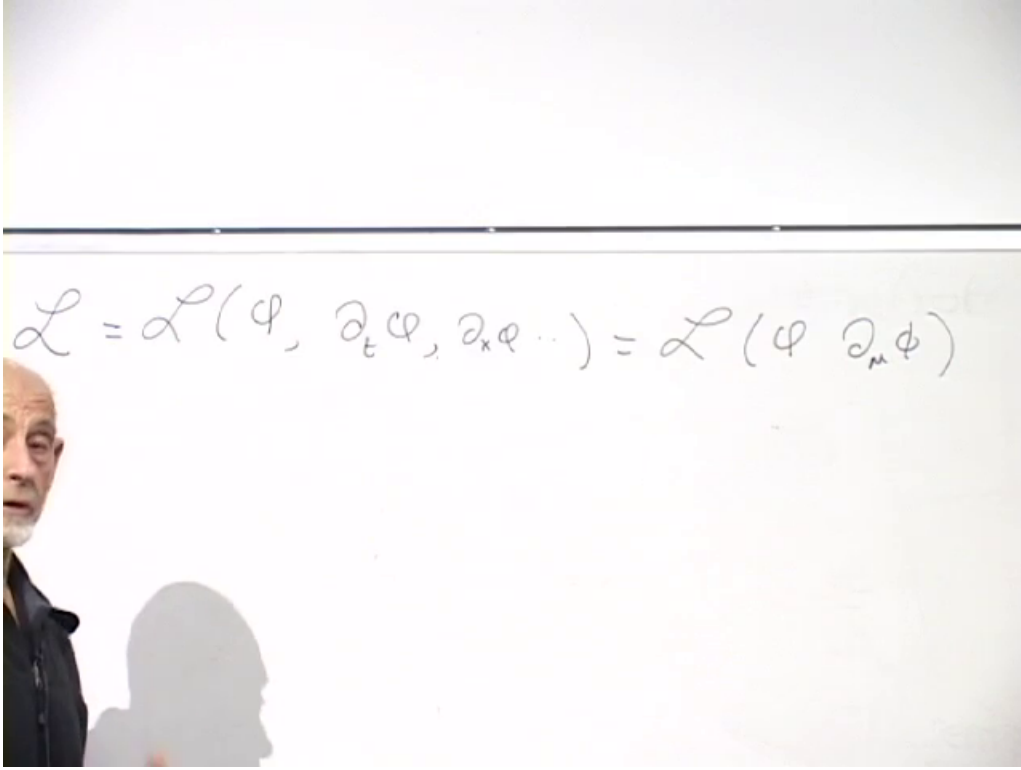


Figure 1.1: Dependence of a scalar-field Lagrangian on the field and its spacetime derivatives.

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The spatial derivative terms carry the opposite sign from the time-derivative term. Differentiating term by term gives

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}, \quad \frac{\partial \mathcal{L}}{\partial \phi_i} = -\phi_i, \quad \frac{\partial \mathcal{L}}{\partial \phi} = -V'(\phi), \quad (1.5)$$

and therefore

$$\ddot{\phi} - \nabla^2 \phi = -V'(\phi). \quad (1.6)$$

This is the equation generated by the scalar Lagrangian.<sup>1</sup>

Here  $V(\phi)$  is the field's potential-energy density. A constant contribution to  $V$  has no effect on this equation because its derivative vanishes. A linear potential is algebraically possible, but it has no lower bound: the field energy can run toward negative infinity. The first stable example is therefore

$$V(\phi) = \frac{1}{2}m^2\phi^2, \quad V'(\phi) = m^2\phi, \quad (1.7)$$

which yields

$$\ddot{\phi} - \nabla^2 \phi = -m^2\phi. \quad (1.8)$$

To expose the meaning of  $m$ , insert a plane wave,

$$\phi(\mathbf{x}, t) = \phi_0 e^{i\omega t - i\mathbf{k} \cdot \mathbf{x}}. \quad (1.9)$$

<sup>1</sup>Editorial clarification: The retained frame is cropped at the left edge; the displayed completion is fixed by the immediately surrounding spoken derivation.

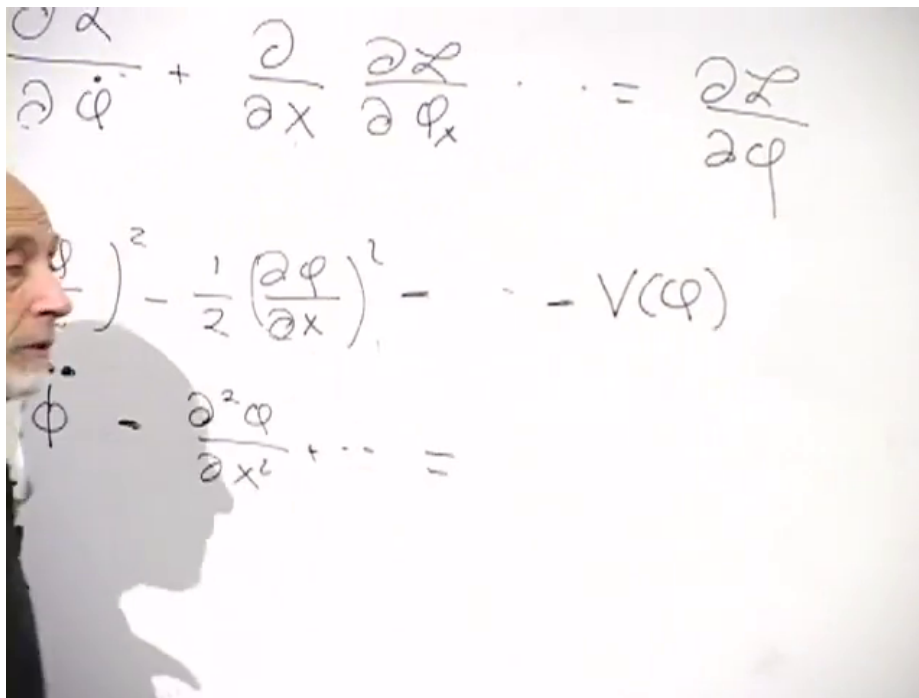


Figure 1.2: Euler–Lagrange equation, scalar-field Lagrangian, and the field equation generated by  $V(\phi)$ .

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Then

$$\ddot{\phi} = -\omega^2 \phi, \quad \nabla^2 \phi = -\mathbf{k}^2 \phi, \quad (1.10)$$

so the wave equation becomes

$$-\omega^2 \phi + \mathbf{k}^2 \phi = -m^2 \phi. \quad (1.11)$$

After canceling  $\phi$ ,

$$\omega^2 = \mathbf{k}^2 + m^2. \quad (1.12)$$

In units with  $\hbar = c = 1$ , frequency is energy and wave number is momentum. Thus

$$E^2 = \mathbf{p}^2 + m^2, \quad (1.13)$$

or, with powers of  $c$  restored,  $E^2 = \mathbf{p}^2 c^2 + m^2 c^4$ . The parameter  $m$  is the mass of one quantum of the field.

Relativistic invariance can be imposed at the level of the compact expression. If  $\mathcal{L}$  is a Lorentz scalar, its equations of motion are Lorentz invariant. The relative sign between temporal and spatial derivatives is the familiar spacetime sign of special relativity.

## 1.2 Nonlinearity and Coupled Classical Fields

Higher powers of a field introduce a new kind of behavior. If

$$V(\phi) = \frac{1}{2} m^2 \phi^2 + g \phi^3, \quad (1.14)$$

then

$$\ddot{\phi} - \nabla^2 \phi = -m^2 \phi - 3g\phi^2. \quad (1.15)$$

Terms quadratic in the Lagrangian give equations linear in the field. Cubic and higher terms give nonlinear equations.

The distinction is visible in the motion of wave packets. For a linear equation, a right-moving packet and a left-moving packet may be added to form another solution. They overlap and interfere for a while, then pass through one another and emerge with their original shapes. With nonlinear terms, the packets influence one another. They can scatter, deform, or break into more complicated waves. The temporary interference of linear waves is not an interaction; a lasting change in the packets signals nonlinearity.

The same pattern appears with two scalar fields. Let

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) + \frac{1}{2}(\partial_\mu \rho)(\partial^\mu \rho) - V(\phi, \rho), \quad (1.16)$$

$$V(\phi, \rho) = \frac{1}{2}m^2 \phi^2 + \frac{1}{2}M^2 \rho^2 + \rho \phi^2. \quad (1.17)$$

Without the last term, the two equations are independent,

$$\ddot{\phi} - \nabla^2 \phi = -m^2 \phi, \quad \ddot{\rho} - \nabla^2 \rho = -M^2 \rho. \quad (1.18)$$

They describe quanta of two species, with masses  $m$  and  $M$ . With the mixed term included,

$$\ddot{\phi} - \nabla^2 \phi = -m^2 \phi - 2\rho \phi, \quad (1.19)$$

$$\ddot{\rho} - \nabla^2 \rho = -M^2 \rho - \phi^2. \quad (1.20)$$

Each equation now contains the other field. The equations are coupled and nonlinear: a  $\phi$  packet scatters a  $\rho$  packet, and conversely. The derivative terms govern the energy–momentum relation, the quadratic undifferentiated terms set the masses, and the mixed higher-order term encodes interaction. This is the basic classical pattern.

### 1.3 Dirac Fields and a Scalar Coupling

For a Dirac field the equation is first order in time. In one-dimensional shorthand, with the extension to three spatial directions understood, its term structure is

$$i\dot{\Psi} \simeq i\alpha \frac{\partial \Psi}{\partial x} + m\beta \Psi, \quad i\alpha \cdot \nabla \Psi = i \sum_i \alpha_i \frac{\partial \Psi}{\partial x_i}. \quad (1.21)$$

The matrices  $\alpha_i$  and  $\beta$  act on the components of  $\Psi$ . A corresponding Lagrangian contains a term linear in  $\partial_t \Psi$ , a spatial-derivative bilinear involving  $\alpha_i$ , and the mass bilinear  $m\Psi^\dagger \beta \Psi$ . Only that structural fact is needed here.<sup>2</sup>

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<sup>a</sup> **Question from the audience.** Is the parenthesis in the attempted Dirac Lagrangian correctly placed, with  $i$  multiplying the whole expression?

**Response.** The placement is not being fixed with confidence here, and it is not needed for the structural point: a Lagrangian exists that generates the Dirac equation.

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<sup>a</sup>Lecture timestamp: 00:26:37.

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<sup>2</sup>Editorial clarification: The signs, parentheses, and placement of  $i$  in the attempted Dirac Lagrangian are explicitly left unsettled in the source. No exact convention-dependent formula is inferred from it.

The image shows a whiteboard with two equations written in black marker. The top equation is the Dirac equation:  $i\dot{\psi} = i\alpha \frac{\partial}{\partial x} \psi + m\beta \psi$ . The bottom equation is the Dirac Lagrangian density:  $i \left( \psi^\dagger \frac{\partial \psi}{\partial t} + \psi^\dagger \alpha \frac{\partial \psi}{\partial x} + \psi^\dagger \beta \psi m \right)$ .

Figure 1.3: Time-derivative, spatial-derivative, and mass terms in the Dirac-field discussion.

*Lecture frame: 00:25:37.*

Now add a scalar field  $\phi$ . The naive product  $\Psi^\dagger \Psi \phi$  is not a Lorentz scalar. Inserting the same Dirac matrix that occurs in the mass term gives

$$\mathcal{L}_{\text{int}} \propto \Psi^\dagger \beta \Psi \phi. \quad (1.22)$$

Equivalently,  $\Psi^\dagger \beta \Psi$  is the scalar Dirac bilinear. Varying the full Lagrangian adds a term proportional to  $\phi$  to the Dirac equation and a source proportional to  $\Psi^\dagger \beta \Psi$  to the scalar equation. The two fields scatter one another.

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<sup>a</sup> **Question from the audience.** Is the  $\beta$  inserted in the scalar–fermion interaction the same  $\beta$  that appears in the Dirac mass term?

**Response.** Yes. It is the same Dirac matrix.

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<sup>a</sup>Lecture timestamp: 00:28:03.

The comparison

$$m\Psi^\dagger \beta \Psi \quad \text{and} \quad \Psi^\dagger \beta \Psi \phi \quad (1.23)$$

makes the coupling look like a mass parameter replaced by a scalar field. This observation will return when a field with a nonzero vacuum value is considered near the end.

So far the interpretation has been classical: wave packets scatter other wave packets. The same algebra has a sharper particle meaning when the fields become operators.

## 1.4 What Powers of Quantum Fields Do

A real scalar quantum field contains both creation and annihilation operators. Schematically,

$$\phi(x) \sim A^+(x) + A^-(x), \quad (1.24)$$

where  $A^+$  creates a quantum and  $A^-$  annihilates one. Consequently,  $\phi^3(x)$  contains four kinds of local process:

$$\phi^3 \sim (A^+)^3 + (A^+)^2 A^- + A^+ (A^-)^2 + (A^-)^3, \quad (1.25)$$

where ordering details and numerical coefficients are suppressed in this schematic reading. The four particle-number changes are

$$3 \longrightarrow 0, \quad 0 \longrightarrow 3, \quad 2 \longrightarrow 1, \quad 1 \longrightarrow 2. \quad (1.26)$$

The three field factors all act at the same point  $x$ . Thus a single cubic term includes three-particle annihilation, three-particle creation, fusion of two quanta into one, and splitting of one quantum into two. In a physical amplitude some of these local readings may later be removed by energy–momentum conservation, but they are all present in the operator product itself.

The quadratic free terms have a different role. First suppress the normalization of the difference between neighboring points  $x$  and  $x'$ :

$$\partial_x \phi \propto \phi(x) - \phi(x'). \quad (1.27)$$

Squaring and expanding gives the term structure

$$\frac{1}{2}(\partial_x \phi)^2 \propto \frac{1}{2}[\phi(x) - \phi(x')]^2 \quad (1.28)$$

$$= \frac{1}{2}\phi(x)^2 + \frac{1}{2}\phi(x')^2 - \phi(x)\phi(x'). \quad (1.29)$$

Each same-point square has several operator readings: it may create two quanta, annihilate two, or annihilate and re-create one at that same point. Those terms do not transport a particle. The cross-term can annihilate a quantum at  $x'$  and create it at  $x$ , or the reverse. It therefore acts as a nearest-neighbor hopping term. Repeated factors carry a particle from  $x$  to  $x'$ , then to  $x''$ , and onward. Quadratic derivative terms govern free propagation; cubic and higher terms scatter, create, and annihilate particles.

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<sup>a</sup> **Question from the audience.** Should the finite-difference representation of the derivative include the separation  $x - x'$ ?

**Response.** Yes. Calling the separation  $L$  supplies the factor  $L^{-2}$  after the difference is squared. That numerical factor changes the coefficient but not the hopping interpretation.

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<sup>a</sup>Lecture timestamp: 00:36:07.

Restoring the separation gives

$$\frac{1}{2}(\partial_x \phi)^2 \longrightarrow \frac{1}{2L^2}[\phi(x) - \phi(x')]^2 \quad (1.30)$$

$$= \frac{1}{2L^2}\phi(x)^2 + \frac{1}{2L^2}\phi(x')^2 - \frac{1}{L^2}\phi(x)\phi(x'). \quad (1.31)$$

The mass term is also quadratic but contains no derivative:

$$\frac{1}{2}m^2\phi(x)^2. \quad (1.32)$$

Its one-particle reading is an incoming quantum absorbed and re-emitted at the same point. It does not move the quantum; it weights, or in that sense counts, its presence there. The same operator square also contains its two-creation and two-annihilation readings. The motion-relevant distinction is therefore between a cross-term joining neighboring points and a mass insertion acting within one point.

## 1.5 The Local Vertex of Quantum Electrodynamics

These elementary particle-process sketches are the beginnings of Feynman diagrams. The central example is quantum electrodynamics, the theory of electrons, positrons, and photons.

The Dirac equation has positive- and negative-energy modes. In the older language,  $\Psi$  contains annihilation operators for both kinds of mode, while  $\Psi^\dagger$  contains the corresponding creation operators. Removing a negative-energy electron is reinterpreted as creating a positive-energy positron. After that relabeling, the useful schematic decomposition is

$$\Psi \sim C_{e-}^- + C_{e+}^+, \quad (1.33)$$

$$\Psi^\dagger \sim C_{e-}^+ + C_{e+}^-. \quad (1.34)$$

Thus  $\Psi$  annihilates an electron or creates a positron, while  $\Psi^\dagger$  creates an electron or annihilates a positron. No normalization, spin label, or momentum integral is meant in this abbreviation.

The QED interaction couples the Dirac current to the electromagnetic vector potential. In the component notation used here, its structure is

$$\mathcal{L}_{\text{QED,int}} \sim e \left( \Psi^\dagger \Psi A_0 + \Psi^\dagger \boldsymbol{\alpha} \Psi \cdot \mathbf{A} \right), \quad (1.35)$$

where

$$J^0 = \Psi^\dagger \Psi, \quad \mathbf{J} = \Psi^\dagger \boldsymbol{\alpha} \Psi. \quad (1.36)$$

The symbol  $\sim$  is important: the relative sign between the temporal and spatial terms depends on the metric and vector-potential conventions. The invariant statement is that the four-current is contracted with the four-potential.<sup>3</sup>

The photon field  $A_\mu$  itself contains photon creation and annihilation operators. Its vector character also carries the photon's polarization—a small directional “flag” in addition to its momentum.

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<sup>a</sup> **Question from the audience.** How can the interaction be a scalar when  $A_\mu$  is a vector and the fermion factor first appears to be a scalar?

**Response.** The fermions supply a four-vector too:  $\Psi^\dagger \Psi$  is its time component and  $\Psi^\dagger \boldsymbol{\alpha} \Psi$  its spatial component. Contracting that current with  $A_\mu$  produces a scalar.

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<sup>a</sup>Lecture timestamp: 00:44:28.

Multiplying the operator contents of  $\Psi^\dagger$ ,  $\Psi$ , and  $A_\mu$  gives four familiar readings of one local vertex:

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<sup>3</sup>Editorial clarification: The source fixes the temporal and spatial components of the current but does not settle a metric convention for the component contraction.

1. an electron emits a photon;
2. an electron absorbs a photon;
3. an electron and a positron annihilate into a photon;
4. a photon creates an electron–positron pair.

For the first reading,  $\Psi$  annihilates the incoming electron,  $\Psi^\dagger$  creates the outgoing electron, and the photon field creates the photon. Choosing the photon-annihilation part gives absorption. Choosing the positron pieces gives pair annihilation or pair creation. One local term contains the whole family.

In a fermion-flow diagram, the arrow on a positron line points opposite to the positron’s direction of propagation. With that convention the fermion arrow passes continuously through every allowed QED vertex. The four readings differ by moving external legs between the incoming and outgoing sides, not by breaking the fermion-flow line.

Every one of these readings carries the same coefficient  $e$ , the electric charge in the microscopic convention being used. This coefficient is the electric charge  $e$ , not the fine-structure constant, although the two are closely related. The vertex amplitude is proportional to  $e$ ; after all kinematic and normalization factors are included, the corresponding probability scale is proportional to  $e^2$ .

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<sup>a</sup> **Question from the audience.** Does the photon-to-electron–positron reading have the same coefficient as the other readings of the vertex?

**Response.** Yes. It carries the same electric-charge coefficient  $e$ .

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<sup>a</sup>Lecture timestamp: 00:49:44.

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<sup>a</sup> **Question from the audience.** Must there be a scaling factor, since a probability has no unit?

**Response.** In the natural microscopic convention being used, the charge  $e$  is dimensionless. It is the amplitude scale, while its square supplies the probability scale.

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<sup>a</sup>Lecture timestamp: 00:49:59.

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A concrete picture is an electron striking a wall and stopping abruptly. There is some probability that energy and momentum leave in the form of a photon. The elementary electromagnetic factor in that radiation amplitude is  $e$ , so its probability scale contains  $e^2$ . This statement isolates the coupling factor; the actual probability also depends on the electron’s state and on the kinematics.

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<sup>a</sup> **Question from the audience.** Does electric charge have units?

**Response.** Not in the natural microscopic convention used here. Macroscopic charge is often reported in coulombs, but particle theory measures charge relative to a chosen elementary unit. In Planck or other natural units the quantities are expressed as pure numbers.

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<sup>a</sup>Lecture timestamp: 00:51:15.

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The contrast is one of conventions and scales. Coulombs are convenient for current in wires; a dimensionless coupling is convenient for microscopic amplitudes.<sup>4</sup>

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<sup>4</sup>Editorial clarification: A brief question from 00:52:37–00:53:15 about “grouping” charge-bearing terms is substantially damaged and elicited no definite construction.

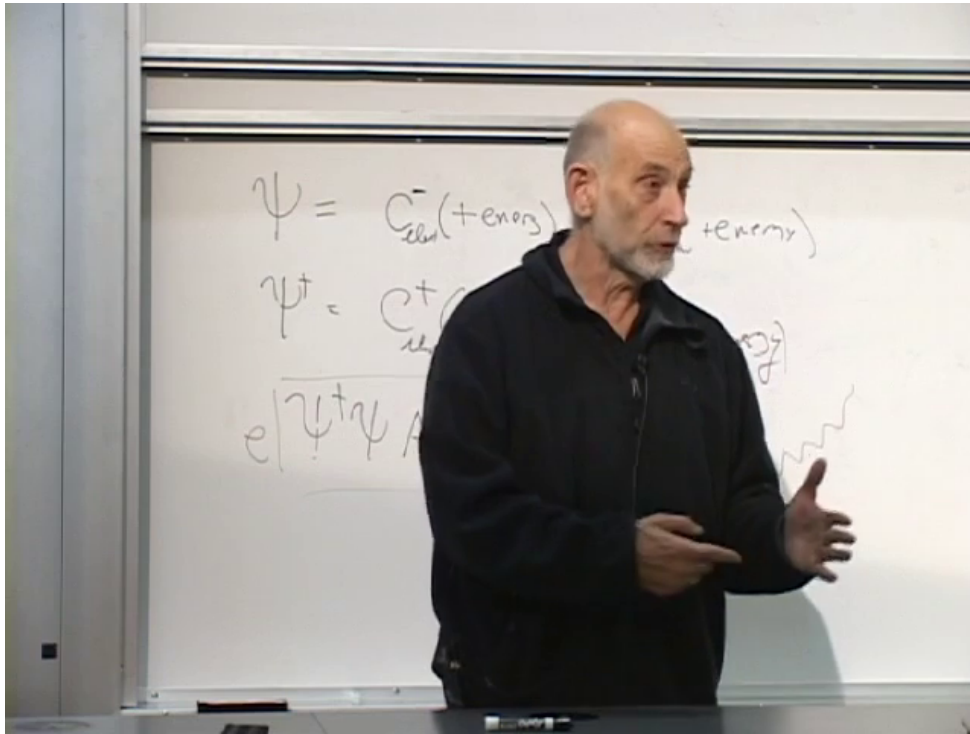


Figure 1.4: Electron and positron operator content of  $\Psi$  and  $\Psi^{\dagger}$ , with an electron–photon interaction sketch.

*Lecture frame: 00:50:39.*

All three fields in the elementary QED interaction are evaluated at one spacetime point. No internal time delay is assigned to the vertex itself.

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<sup>a</sup> **Question from the audience.** Are the absorption and re-emission of the electron and the emission of the photon simultaneous, or does the equation build in a delay?

**Response.** At the elementary vertex all the operators are evaluated at the same spacetime point. The basic process therefore contains no delay between those acts.

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<sup>a</sup>Lecture timestamp: 00:53:19.

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<sup>a</sup> **Question from the audience.** Must each conserved quantity be conserved separately?

**Response.** Yes. Electric charge is one of the quantities that must be conserved separately.

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<sup>a</sup>Lecture timestamp: 00:53:57.

Charge conservation is visible both in the diagram and in the algebra. A process that turned an incoming electron into an outgoing positron would reverse the charge and is not an allowed QED vertex. With time drawn upward, an electron arrow follows time upward and a positron arrow points downward; the valid vertex always has a through-going fermion arrow. Algebraically, the interaction contains one  $\Psi$  and one  $\Psi^{\dagger}$ . The first raises charge by one unit when it annihilates an electron or creates a positron, and the second lowers it by one unit. Their effects cancel.

<sup>a</sup> **Question from the audience.** Is the boxed electron–photon expression the whole QED Lagrangian?

**Response.** No. It is the interaction term. The quadratic free-photon terms and the quadratic Dirac terms, which govern propagation, have not been written.

<sup>a</sup>Lecture timestamp: 00:56:05.

<sup>a</sup> **Question from the audience.** Is a classical Lagrangian deduced while a quantum Lagrangian is guessed and tested, and does the kinetic-minus-potential structure remain in quantum mechanics?

**Response.** The kinetic-minus-potential pattern remains: time derivatives provide the kinetic form, spatial gradients generalize it, and undifferentiated field terms provide the potential form. The quantum Lagrangian is the more fundamental description. In a state containing very many quanta, the same Lagrangian produces the classical wave equations. Candidate Lagrangians are proposed using physical principles and deeper theory, then established or rejected through collision experiments.

<sup>a</sup>Lecture timestamp: 00:56:35.

The large-occupation limit explains why the same expression can govern a quantum particle theory and its classical waves. This agreement is not a separate miracle imposed afterward; the classical field is the collective behavior of many quanta. Conversely, experimental collisions expose the primitive vertices directly enough to determine the QED and Standard Model interactions with great precision.

## 1.6 Crossing and Spacetime Conservation Laws

The legs of a local vertex may be crossed between incoming and outgoing positions. An outgoing particle moved to the incoming side is reinterpreted as its antiparticle, and conversely. In this sense the same electron–positron–photon operator has emission, absorption, pair-annihilation, pair-creation, and even an all-incoming local reading.

<sup>a</sup> **Question from the audience.** If a basic vertex diagram is rotated, does it remain a valid local reading?

**Response.** Yes. Its external legs may be moved between incoming and outgoing positions, with the corresponding particle–antiparticle reinterpretation.

<sup>a</sup>Lecture timestamp: 00:59:34.

The local operator reading is not yet a complete physical process. Its amplitude must be integrated over every spacetime point at which the vertex could occur. The time integral enforces energy conservation and the space integral enforces momentum conservation. A picture in which an electron, a positron, and a photon all disappear into the vacuum is present as a formal local reading, but the integrated amplitude vanishes because the process cannot conserve energy and momentum.

<sup>a</sup> **Question from the audience.** Does the all-incoming picture mean that an electron, a positron, and a photon can disappear into nothing?

**Response.** It is a reading of the local operator, but the physical amplitude includes integration over the vertex position. Those integrals impose energy and momentum conservation, so this kinematically forbidden process has zero amplitude.

<sup>a</sup>Lecture timestamp: 01:00:30.

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<sup>a</sup> **Question from the audience.** Does the converse vacuum-to-electron-positron-photon reading exist, and do excluded processes disappear after integration?

**Response.** The converse local reading is also present. Integration over space and time removes any process incompatible with the conservation laws.

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<sup>a</sup>Lecture timestamp: 01:01:57.

Derivatives in the Lagrangian fit the operator expansion without differentiating the creation and annihilation symbols themselves. A field has the schematic form

$$\Psi(x) \sim \int d^3k [C(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{x}-i\omega t} + \dots]. \quad (1.37)$$

The operators  $C(\mathbf{k})$  are coefficients in this spacetime expansion. Derivatives act on the plane waves, pulling down  $\mathbf{k}$  or  $\omega$ .

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<sup>a</sup> **Question from the audience.** When differentiating a field expansion, is the creation operator itself differentiated?

**Response.** No. The creation and annihilation operators are independent of the spacetime coordinate in this expansion. The derivatives act on the exponential factors and bring down momentum and frequency.

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<sup>a</sup>Lecture timestamp: 01:02:27.

## 1.7 Nucleons and Pions as an Effective Theory

The same language can describe protons, neutrons, and pions. Protons and neutrons are fermions represented by fields  $\Psi_p$  and  $\Psi_n$ . For this interaction-counting discussion the pion fields are treated as scalar fields in three charge varieties,

$$\pi^+, \quad \pi^-, \quad \pi^0. \quad (1.38)$$

The proton and  $\pi^+$  carry one positive unit of electric charge, the  $\pi^-$  carries one negative unit, and the neutron and  $\pi^0$  are neutral. At a deeper level pions are quark–antiquark systems, but that internal description can initially be left unresolved.

A schematic charge-conserving list of nucleon–pion field products is

$$\mathcal{L}_{N\pi} \sim \left\{ \Psi_p^\dagger \Psi_n \pi^+, \Psi_n^\dagger \Psi_p \pi^-, \Psi_p^\dagger \Psi_p \pi^0, \Psi_n^\dagger \Psi_n \pi^0 \right\}. \quad (1.39)$$

Only the field content and charge assignments are being displayed; relative coefficients, signs, and the spinor-matrix structure are not fixed here. The charge on the first pion field is fixed by testing a concrete reading: a neutron together with a positive pion may produce a proton, whereas using a negative pion there would turn negative incoming charge into positive outgoing charge. The conjugate term gives the opposite charged process.<sup>5</sup>

Each term again has several crossed readings. Moving an outgoing neutron to the incoming side changes it into an incoming antineutron. The charged vertex can therefore describe a proton and an antineutron combining into a  $\pi^+$ , as well as processes such as  $p \rightarrow n + \pi^+$  when the kinematics

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<sup>5</sup>Editorial clarification: The pion sign is corrected several times in the source before the charge-conserving pair shown here is reached; the displayed ordering records that final schematic structure.

permit. The neutral terms let a proton or neutron emit or absorb a  $\pi^0$ . Charge conservation selects the allowed family, while energy, momentum, and any further symmetries determine which member can occur in a particular experiment.

This was a useful effective description before quark substructure was known. It remains useful whenever a probe cannot resolve that structure. A proton has a finite distribution of constituents, rather as an atom has internal structure, although on a much smaller scale. If an experiment transfers too little momentum to break up or resolve the proton, the proton may be treated as a point particle with its own field. A sufficiently hard collision exposes the quarks, and the point-particle proton field is then no longer adequate.

Because the proton is charged, it also has a local proton–photon interaction, schematically  $J_p^\mu A_\mu$ , where  $J_p^\mu$  is the proton current. The electron has its corresponding electron–photon interaction. A proton cannot instead turn into a neutron by emitting a photon:  $p \rightarrow n + \gamma$  fails charge conservation.

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<sup>a</sup> **Question from the audience.** What happens when a proton and an electron interact?

**Response.** Each charged particle has a photon vertex. The proton can emit a photon while remaining a proton, and the electron can absorb the photon while remaining an electron. Proton–electron scattering is assembled from those two local vertices and propagation of the photon between them.

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<sup>a</sup>Lecture timestamp: 01:10:28.

## 1.8 Locality and Repeated Elementary Processes

There is no special nonlocal term for the complete proton–electron exchange. One local vertex emits the photon at the proton, quadratic photon terms carry it from neighboring point to neighboring point, and another local vertex absorbs it at the electron. The extended process is a product of those elementary pieces.

$$[J_p^\mu A_\mu](x_p) \longrightarrow \underbrace{x_p \rightarrow x_1 \rightarrow x_2 \rightarrow \cdots \rightarrow x_e}_{\text{quadratic photon propagation}} \longrightarrow [J_e^\nu A_\nu](x_e). \quad (1.40)$$

The arrows in this schematic chain denote successive neighboring-point insertions, not a new nonlocal interaction.

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<sup>a</sup> **Question from the audience.** How would one write the Lagrangian for the whole proton–electron exchange?

**Response.** One does not add a term for the whole extended event. It is built from a local proton–photon vertex, repeated quadratic propagation terms, and a local electron–photon vertex.

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<sup>a</sup>Lecture timestamp: 01:12:06.

The local structure has two forms. An undifferentiated interaction term acts at one spacetime point. A derivative term, expressed as a finite difference, connects that point only with a neighboring point. Neither kind transfers a particle directly across a finite distance. Long-range propagation is assembled from many infinitesimal moves; there is no elementary action at a distance.

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<sup>a</sup> **Question from the audience.** Does the Lagrangian contain terms only at one point?

**Response.** Interaction terms act at one point. Derivative terms connect one point with an infinitesimally neighboring point. Emission, many neighboring hops, and absorption together produce a finite exchange.

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<sup>a</sup>Lecture timestamp: 01:14:36.

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<sup>a</sup> **Question from the audience.** Are all the pieces of the extended process present, and does compounding them mean multiplication?

**Response.** Yes. Each elementary emission, hop, or absorption is encoded in the Lagrangian. Products of repeated insertions compound those pieces into the larger process.

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<sup>a</sup>Lecture timestamp: 01:15:36.

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The relativistic quantum field theory under discussion presumes the spacetime setting on which its fields are defined. It should not be confused with a complete quantum theory of a dynamical spacetime geometry.

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<sup>a</sup> **Question from the audience.** Does this compounding remain valid in a strongly relativistic setting such as near a black-hole horizon?

**Response.** The black-hole extension is outside the scope of this construction. The quantum field theory being used here does not by itself provide a complete description of black-hole dynamics, where gravity is essential.

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<sup>a</sup>Lecture timestamp: 01:16:28.

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A field is an operator for an elementary local act. One field factor can absorb a particle or emit one. A product of two factors can absorb one particle and emit another at the same point; higher products can change the particle content further. Products of local operators join these acts into histories. In spacetime, motion is represented by absorption in one small region, re-emission into a neighboring region, another absorption, and another re-emission. Taking the step size toward zero while the number of steps grows gives the continuum trajectory.

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<sup>a</sup> **Question from the audience.** How should fields, distance, and time be understood in this local-step picture?

**Response.** Fields are operators for local absorption and emission. Products join those elementary acts, and motion through spacetime is built from infinitesimal neighboring steps before the continuum limit is taken.

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<sup>a</sup>Lecture timestamp: 01:17:30.

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A regulated starting point is therefore a division of spacetime into small cells, with the field operators attached to the cells.<sup>6</sup> Derivatives become differences. Quadratic cross-terms connect neighboring cells, while mass and interaction terms act inside one cell. The cells are a device for defining the limiting theory, not additional particles or a new physical mass scale.

Special-relativistic effects need not be added separately at every step. When the Lagrangian is a Lorentz scalar, the amplitudes assembled from it are Lorentz invariant, and effects such as time dilation follow from that structure.

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<sup>6</sup>Editorial clarification: The audience wording at 01:17:30–01:20:40 is partly obscured; the paraphrased question preserves only its recoverable subject, while the response about local operators and successive spacetime steps is clear.

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<sup>a</sup> **Question from the audience.** Do high-energy motion and time dilation create extra problems in a multistep process?

**Response.** No separate correction is required. If the Lagrangian is a Lorentz scalar, special-relativistic effects such as time dilation are already built into the result. The precise concern behind the question is otherwise not fully clear.

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<sup>a</sup>Lecture timestamp: 01:21:10.

Creation and annihilation operators may be represented in a position basis or in a momentum basis. A position-space creator is a Fourier superposition of momentum-space creators, and conversely. There is no operator that creates a state with both exactly specified position and exactly specified momentum. Localizing the particle sharply requires a wide band of momenta; localizing it sharply in momentum spreads it over position. The Fourier relation between the two bases is the uncertainty principle in this language.

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<sup>a</sup> **Question from the audience.** How are creation and annihilation operators related to position, momentum, and the uncertainty principle?

**Response.** They can be represented as functions of position or as functions of momentum. The two representations are Fourier transforms of one another. A sharply localized position state requires many momenta, while a sharply localized momentum state is spread over position; there is no creator for simultaneously sharp  $x$  and  $p$ .

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<sup>a</sup>Lecture timestamp: 01:22:48.

The cell construction can now be stated completely. Label the cells by  $n$ , replace a derivative by a finite difference such as

$$\partial_\mu \phi(x) \longrightarrow \frac{\phi_{n+\hat{\mu}} - \phi_n}{a}, \quad (1.41)$$

and place every field species in every cell. A quadratic cross-term removes a quantum from one cell and creates it in a neighbor. The continuum theory is obtained by taking the spacing  $a$  toward zero after the regulated expressions have been defined.

$$\underbrace{\phi_n \phi_{n+\hat{\mu}}}_{\text{hop } n \leftrightarrow n+\hat{\mu}}, \quad \underbrace{m^2 \phi_n^2}_{\text{same-cell mass insertion}}. \quad (1.42)$$

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<sup>a</sup> **Question from the audience.** Can the subdivision of spacetime into cells be completed?

**Response.** Yes. The fields are indexed by cells, derivatives become finite differences, and quadratic terms hop particles between neighboring cells. The continuum is recovered as the cell size tends to zero.

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<sup>a</sup>Lecture timestamp: 01:25:16.

Electrons hop between cells through their quadratic terms, and photons do the same. When an electron and photon meet in one cell, the QED interaction may absorb and re-emit the electron while creating or annihilating the photon. The scalar mass term has a same-cell reading: a particle enters and leaves that cell with a coefficient proportional to  $m^2$ . It does not supply a hop.

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<sup>a</sup> **Question from the audience.** Is the cell size related to the mass coefficient?

**Response.** No. The cell size is a regulator spacing that is ultimately taken to zero; the mass coefficient is an independent parameter in the Lagrangian.

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<sup>a</sup>Lecture timestamp: 01:28:18.

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<sup>a</sup> **Question from the audience.** Is multiplying the local factors a joint-amplitude construction over spacetime rather than merely a sequence in time?

**Response.** Yes. The cells cover spacetime, and their local contributions combine across that spacetime grid. The product supplies a joint amplitude for the required local events and propagation steps.

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<sup>a</sup>Lecture timestamp: 01:28:43.

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## 1.9 Sources, Detectors, and the Dynamics Between Them

A complete process is a history in spacetime. Its initial condition is fixed by what the experimenter prepares, and its final condition by what the detectors register. A simple source is a hot filament that emits an electron. A field operator at the filament creates that incoming electron; field operators at detector positions describe its eventual absorption.

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<sup>a</sup> **Question from the audience.** What is the initial stage of such a spacetime process?

**Response.** It is prepared at the sources. An electron emitted by a hot filament is a simple example: a field operator at the filament creates the electron, while field operators at the detectors specify the final absorption events.

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<sup>a</sup>Lecture timestamp: 01:29:30.

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The same construction can be stated from the vacuum. Creation operators at the sources build the incoming state on the vacuum. Annihilation operators associated with the detectors select the outgoing state. These operators specify boundary conditions; they do not replace the dynamics throughout the region between source and detector.

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<sup>a</sup> **Question from the audience.** Would it be wrong to place the dynamics only at the emitters rather than throughout the system?

**Response.** Yes. Start with the vacuum, use source operators to prepare the incoming particles, and use detector operators to select the final particles. Between them, repeated Lagrangian insertions act throughout spacetime, moving particles, creating them, and annihilating them.

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<sup>a</sup>Lecture timestamp: 01:31:11.

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With enough insertions, several particles can move simultaneously through many neighboring cells; particles can also be created or annihilated along the way. The initial operators say what enters, the final operators say what is registered, and the repeated Lagrangian operations supply everything between them.

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<sup>a</sup> **Question from the audience.** If arbitrarily many Lagrangian insertions can occur, are their positions being integrated over?

**Response.** Yes. Their spacetime positions are integrated over. The exact Feynman rules are not being derived here; the immediate point is that simple local field expressions encode the primitive interactions from which a full process is assembled.

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<sup>a</sup>Lecture timestamp: 01:33:24.

## 1.10 Phase Symmetry, Charge, and Amplitudes

Writing the primitive processes in a Lagrangian also makes their symmetries visible. Consider a global phase change of the electron field,

$$\Psi \longrightarrow e^{i\theta}\Psi, \quad \Psi^\dagger \longrightarrow e^{-i\theta}\Psi^\dagger, \quad A_\mu \longrightarrow A_\mu. \quad (1.43)$$

A term with equal numbers of  $\Psi$  and  $\Psi^\dagger$  is unchanged because the phases cancel. A term with unequal numbers would acquire a phase. The equality of those numbers is the algebraic form of charge conservation.

This is a central advantage of the Lagrangian description. Its field factors display the elementary processes, their combinations display the symmetries, and the symmetries display the conservation laws. Repeated insertions evolve the prepared initial state into the selected final state. The result is a quantum amplitude. Exact evaluation requires the Feynman rules, but the structural content is already visible: local operations repeat in every order allowed by the conservation laws.

## 1.11 Field Measurements and Commutators

Field operators retain the usual quantum-mechanical meaning of operators. Their eigenvalues are possible results of field measurements, and their eigenvectors describe states associated with those values. An electric field can be probed by placing a charged particle in it and observing how the particle accelerates.

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<sup>a</sup> **Question from the audience.** Do the eigenvalues and eigenvectors of field operators play their usual quantum-mechanical role?

**Response.** Yes. Eigenvalues are possible measured field values, and fields can be probed through their effects on particles. Commutators determine the restrictions on simultaneous measurements.

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<sup>a</sup>Lecture timestamp: 01:37:06.

The trivial identity

$$[\phi(x), \phi(x)] = 0 \quad (1.44)$$

does not settle the measurement question. One must specify two observables and their spacetime relation. A field and its conjugate momentum form a canonical noncommuting pair, and field operators at timelike-related points need not commute; local bosonic fields at spacelike separation do commute, as locality requires. The familiar connection between commutators and uncertainty therefore carries over, but only after the operator pair has been identified correctly.<sup>7</sup>

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<sup>7</sup>Editorial clarification: The source states noncommutation at distinct points too broadly. The qualification here preserves its measurement point while enforcing the microcausality of spacelike-separated local bosonic fields.

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<sup>a</sup> **Question from the audience.** Is the electric or electromagnetic field measured in volts, or in volts per meter?

**Response.** The electric field is measured in volts per meter. Volts and meters are not the most useful language for microscopic field processes, which are more naturally organized in terms of photons. Conversely, quantum field theory is not the useful language for deciding how to water a house.

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<sup>a</sup>Lecture timestamp: 01:39:57.

The units natural to one description need not be the illuminating units in another. Volts and volts per meter fit macroscopic apparatus; photon creation and annihilation fit microscopic processes.

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<sup>a</sup> **Question from the audience.** Is the central idea that Lagrangian terms are building blocks combined into larger processes?

**Response.** Yes. Elementary processes repeat from one point to the next. Each basic element acts at one spacetime point, or connects neighboring points through a derivative term. Apparently nonlocal propagation is built from these local steps; there is no elementary action at a distance.

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<sup>a</sup>Lecture timestamp: 01:41:10.

## 1.12 Products of Local Processes and Sums over Routes

There is a compact heuristic way to display the repetition of local processes. Divide spacetime into cells and write

$$\prod_x (1 + \mathcal{L}_I(x)), \quad (1.45)$$

where  $x$  labels the cells and  $\mathcal{L}_I(x)$  is the local interaction Lagrangian. This is not the exact continuum formula. It is a combinatorial stand-in that exposes how products generate every allowed selection of local events.

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<sup>a</sup> **Question from the audience.** Can many creation, propagation, and destruction steps be compacted into one expression?

**Response.** Heuristically, multiply one factor  $1 + \mathcal{L}_I(x)$  for every spacetime cell. Expanding the product generates no event, one event at any selected point, or events at several points.

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<sup>a</sup>Lecture timestamp: 01:42:18.

For two cells,

$$(1 + \mathcal{L}_I(x))(1 + \mathcal{L}_I(x')) = 1 + \mathcal{L}_I(x) + \mathcal{L}_I(x') + \mathcal{L}_I(x)\mathcal{L}_I(x'). \quad (1.46)$$

The first term means that neither interaction occurs. The next two place one elementary interaction at  $x$  or at  $x'$ . The last term places interactions at both points. With many factors there are terms involving one point, two points, three points, and so on. A term with several vertices can describe, for example, two incoming particles producing three outgoing particles.

Quadratic propagation pieces enter the same construction. A finite displacement is represented by a chain of neighboring insertions. Somewhere in the expanded product are factors at one cell, its neighbor, the next neighbor, and onward, carrying a particle between the endpoints.

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<sup>a</sup> **Question from the audience.** If finite propagation requires many local motion terms, how is the required motion recovered?

**Response.** The large product contains chains of neighboring insertions that carry the particle from its initial point to its final point. There are generally many such chains along different routes. Each route contributes an amplitude, and the total amplitude is their sum.

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<sup>a</sup>Lecture timestamp: 01:46:29.

If  $r$  labels a route, then schematically

$$\mathcal{A}_{\text{total}} = \sum_r \mathcal{A}_r, \quad P = |\mathcal{A}_{\text{total}}|^2. \quad (1.47)$$

Ordinary probabilities are not assigned to the routes and then added. The amplitudes are added first:

$$\left| \sum_r \mathcal{A}_r \right|^2 = \sum_r |\mathcal{A}_r|^2 + \sum_{r \neq s} \mathcal{A}_r \mathcal{A}_s^*. \quad (1.48)$$

The cross-terms are interference.

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<sup>a</sup> **Question from the audience.** Is this consistent with the double-slit experiment?

**Response.** Exactly. The alternatives through the two slits contribute amplitudes. Squaring their sum produces interference terms, just as the many spacetime routes in the local-step construction interfere.

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<sup>a</sup>Lecture timestamp: 01:48:27.

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<sup>a</sup> **Question from the audience.** Where does the 1 in each factor  $1 + \mathcal{L}_I$  come from?

**Response.** It is a combinatorial device that lets a cell contribute no interaction as well as an interaction. It is related to path-integral methods, but the displayed product is only a heuristic approximation to the technical expression.

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<sup>a</sup>Lecture timestamp: 01:49:00.

The exact formalism supplies the proper spacetime integrations, ordering, normalization, and phases. The product is useful because it reveals the combinations without pretending to be the full formalism.

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<sup>a</sup> **Question from the audience.** Is this the principle of stationary action?

**Response.** No. Classical stationary action selects a distinguished classical trajectory. Quantum mechanics instead sums amplitudes over trajectories, with their weights governed by the action. Even an extremely indirect route—going through Paris on the way from an office to a nearby house—belongs to the quantum sum.

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<sup>a</sup>Lecture timestamp: 01:49:57.

The contrast is simple even though the relation is subtle. Classical mechanics selects a stationary path. Quantum mechanics sums paths; in the appropriate limit, phases reinforce around the classical trajectory.

### 1.13 Inferring the Lagrangian and Choosing a Sector

Once a Lagrangian and a specified process are given, the rules for computing its probability are definite. A calculation may still demand ingenuity, as difficult mathematics often does, but the deeper uncertainty lies in discovering which fields and couplings nature uses.

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<sup>a</sup> **Question from the audience.** Is the art in expressing the Lagrangian or in using it?

**Response.** The main art is inferring the Lagrangian from experimental data. Once it is known, a process such as two particles entering and four leaving has a definite amplitude to be calculated by precise rules. Carrying out the mathematics can require cleverness, but that is not an ambiguity in the theory.

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<sup>a</sup>Lecture timestamp: 01:50:43.

Particle collisions expose the elementary vertices. Terms are proposed, often under the guidance of symmetry and deeper principles, and the resulting amplitudes are tested against the observed outcomes. Success fixes both the allowed combinations of fields and their numerical coefficients.

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<sup>a</sup> **Question from the audience.** Is there a separate Lagrangian for every interaction—a large book of Lagrangians?

**Response.** The working picture is instead one large Lagrangian containing all the fields and interactions, although that picture may ultimately be false. Recognizable sectors occur as pieces of the sum. A new particle adds a field; a new primitive process adds an interaction term.

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<sup>a</sup>Lecture timestamp: 01:52:24.

The complete expression can be untidy without being book-length. As a deliberately rough scale, one may imagine of order a hundred objects called elementary particles and several hundred terms. Quantum electrodynamics, the strong and weak interactions, and a gravitational sector could still occupy only a few densely written pages. The exact count is not the point. The working picture is one dynamical expression assembled from many recognizable pieces, and it is explicitly a picture that may eventually prove incomplete.

Whether the fields determine a more fundamental spacetime geometry lies beyond the standard quantum-field framework being used here.<sup>8</sup>

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<sup>a</sup> **Question from the audience.** Are the experimentally determined numbers the coupling constants?

**Response.** Coupling constants are coefficients of terms in the Lagrangian. The coefficient  $m^2$  of a quadratic mass term is one example; coefficients of  $\phi^4$ ,  $\phi^6$ , or still higher powers are others.

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<sup>a</sup>Lecture timestamp: 01:55:22.

One coefficient controls a family of related amplitudes because each field product has several creation and annihilation readings. Knowing the Lagrangian therefore determines far more than one isolated reaction.

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<sup>8</sup>Editorial clarification: The question at 01:55:00–01:55:20 is too damaged to identify the manifold or geometric construction reliably; only this boundary-of-scope response is secure.

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<sup>a</sup> **Question from the audience.** Must the entire Lagrangian be known in order to calculate quantum electrodynamics?

**Response.** Infinite precision would require every sector through which electrons and photons can interact. At atomic energies, however, the electron–photon sector can normally be isolated. There is not enough energy to resolve tiny quark structure, so other sectors can be neglected to the required accuracy.

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<sup>a</sup>Lecture timestamp: 01:56:05.

Practical calculations use the sector relevant to the experimental scale. This resembles hydrodynamics: a complicated microscopic system can admit a precise coarse-grained description without displaying all its deeper constituents. Quantum field theory can therefore be phenomenological and exact at once—phenomenological in the fields and coefficients inferred from experiment, exact in the rules applied after those ingredients are specified.

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<sup>a</sup> **Question from the audience.** Does the Standard Model include gravity, and do these relativistic Lagrangians include general relativity?

**Response.** The theory officially called the Standard Model does not include gravity, although gravity may be appended when discussing a broader theory. The field theories considered here combine quantum mechanics with special relativity. They do not by themselves include general relativity.

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<sup>a</sup>Lecture timestamp: 01:58:09.

## 1.14 A Vacuum Value as a Mass Coefficient

The mass of a scalar field appears as the coefficient of  $\phi^2$ . An interaction with another field can therefore imitate a mass term when that other field has a nonzero vacuum value.

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<sup>a</sup> **Question from the audience.** If particle masses arise through Higgs interactions, can a Higgs coupling function as a mass term?

**Response.** Yes. A term  $H\phi^2$  is cubic and ordinarily represents an interaction with two  $\phi$  legs and one  $H$  leg. If the potential  $V(H)$  selects a vacuum with nonzero  $H$ , that vacuum value is a number multiplying  $\phi^2$ , so it contributes an effective quadratic mass term.

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<sup>a</sup>Lecture timestamp: 01:59:16.

Consider schematically

$$\mathcal{L} \supset H\phi^2 - V(H). \quad (1.49)$$

As an operator term,  $H\phi^2$  can absorb a  $\phi$ , emit a  $\phi$ , and emit or absorb an  $H$  quantum. Suppose the potential has two minima away from  $H = 0$ , and the vacuum selects one of them:

$$\langle H \rangle = v \neq 0. \quad (1.50)$$

Write the field as its vacuum value plus a fluctuation,

$$H = v + h. \quad (1.51)$$

Then

$$H\phi^2 = v\phi^2 + h\phi^2. \quad (1.52)$$

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The first term is quadratic in  $\phi$  and contributes to its mass coefficient; the second remains an interaction with the fluctuating field  $h$ .<sup>9</sup>

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<sup>a</sup> **Question from the audience.** Does the Standard Model include virtual-particle creation and annihilation?

**Response.** Absolutely. Virtual creation and annihilation are intrinsic to the repeated quantum processes encoded by the Standard Model Lagrangian.

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<sup>a</sup>Lecture timestamp: 02:01:29.

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<sup>9</sup>Editorial clarification: The source ends the sentence while stating that the nonzero vacuum value becomes a mass coefficient. The displayed substitution is the immediate algebraic completion; its normalization and sign depend on the full Lagrangian conventions.